

Westly Bouchard

Denver, CO | 720-418-0939

westlybouchard.com | westly.bouchard@gmail.com | linkedin.com/in/wcbouchard

Projects

Open Holonomics | ROS2, C++, Python, Onshape, Manufacturing May 2025 - Present

- Designed and manufactured a compact, open-source, swerve drive module, utilizing custom gear reductions to fit within a constrained 12x12-inch footprint.
- Developed a modular, node-based ROS2 control stack to handle forward kinematics and command generation.
- Iterated through multiple rapid prototyping cycles to optimize manufacturability, fine-tuning gear tooth tolerances and bearing press-fits to achieve consistent dimensional accuracy across four identical modules.

Motor Driver Interface | C++, ROS2_control, Linux, CAN Bus May 2025 - August 2025

- Developed a custom C++ `ros2_control` hardware interface to communicate with legacy ODrive motor controllers over a CAN bus network.
- Implemented asynchronous, non-blocking CAN frame read/write operations utilizing Linux SocketCAN and the `epoll()` syscall, ensuring deterministic executing without thread-spinning.
- Modified legacy open-source motor firmware to multiplex dual-encoder inputs over the bus, significantly reducing system wiring complexity.

Encoder PCB | KiCad, PCB Design, AS5047P, SPI June 2025 - December 2025

- Designed and routed a custom rotary magnetic position sensor PCB in KiCad based on the AS5047p IC for use in robotic actuators.
- Engineered the board to support both 3.3V and 5V logic and successfully broke out SPI, ABI, and UVB communication interfaces for maximum hardware compatibility.

Experience

Software Engineer Intern | Freewheel (Beeswax App Team) Jun 2025 - Aug 2025

- Contributed to the development of a large, technically complex web application.
- Rewrote outdated PHP APIs using Python and Django to modernize backend infrastructure.
- Ensured strict interoperability between legacy codebases and the modern front-end, working extensively with React.js, Typescript, MySQL, Postgres, and Snowflake.
- Managed version control and agile workflows using Git, GitHub, and Scrum.

Computer Science Teaching Assistant | Colorado School of Mines Jan 2024 - Present

- Guided engineering students through their foundational C++ coursework.
- Supported student comprehension of complex memory management and object-oriented topics, including pointers, inheritance, and polymorphism.

Education

Colorado School of Mines | Golden, CO Aug 2023 - May 2027

- **Bachelor of Science in Design Engineering**
- **Concentration:** Robotics & Automation | **Minor:** Computer Science
- **GPA:** 3.9 / 4.0 (Dean's List)

Skills

- **Software & Tools:** C++, Python, ROS2, Linux, Git / GitHub, Django, React, Machine Learning
- **Electrical & Hardware:** KiCad (PCB Design), Verilog, Logic Design, SPI / CAN Protocols, Embedded C
- **Mechanical & Design:** SolidWorks (CSWP), Onshape, 3DP / DFM, Prototyping, Feedback Control Analysis